Answer ALL questions. Write your answers in the spaces provided.

1. $f(x) = x^4 + x^3 + 2x^2 + ax + b,$

where a and b are constants.

When $f(x)$ is divided by $(x - 1)$, the remainder is 7

(a) Show that $a + b = 3$

(2)

When $f(x)$ is divided by $(x + 2)$, the remainder is -8

(b) Find the value of a and the value of b

(5)